

Yingjie Lei

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Education

- Johns Hopkins University** Aug. 2026 – Present
M.S.E. in Computer Science
- South China Normal University – Aberdeen Institute of Data Science & AI** Sep 2022 – Jun 2026
B.Eng. & B.Sc. (Hons) in Artificial Intelligence
- CN Degree: B.Eng. in Artificial Intelligence, South China Normal University (Project 211)
 - UK Degree: B.Sc. (Hons) in Artificial Intelligence, University of Aberdeen; graduated with First Class Honours
 - **GPA:** 90.9/100 (CN), 20.23/22 (UK); **Rank:** 5/88 (Top 5.7%)

Research Interests

Embodied AI, robot learning, and representation learning.

Publications & Preprints

- PrefBench: Evaluating Zero-Shot LLM Agents in Hidden-Preference Personalized Pricing Negotiations** [🔗](#) May 2026
Yingjie Lei
 arXiv preprint, arXiv:2605.22855, 2026.
- Draft-and-Target Sampling for Video Generation Policy** [🔗](#) Under Review
 Qikang Zhang, **Yingjie Lei**, Wei Liu, Daochang Liu
 Under review, 2026.
- Test Time Training for 4D Medical Image Interpolation** [🔗](#) Apr. 2025
 Qikang Zhang*, **Yingjie Lei***, Zihao Zheng, Ziyang Chen, Zhonghao Xie
 IJCNN 2025, Oral Presentation. *Equal contribution.

Research Experience

- Representation-to-Decision Analysis in LeWM** Jun. 2026 – Sep. 2026
Independent Research
- Designed a fixed-LeWM case study across Reacher, TwoRoom, Cube, and PushT, comparing predicted and realized outcomes for matched action candidates.
 - Traced physical-state and command interventions through representation, prediction, goal-distance cost, candidate selection, and executed actions.
 - Localized task-specific mismatches: Cube action differences nearly ignored by the encoder changed block motion; Reacher hidden velocities changed future motion; TwoRoom execution-equivalent commands changed predictions.
 - In PushT, decomposed goal-distance costs using physically edited references and found score interactions that could reinforce or offset wrong candidate preferences.
 - Tested rescoring and elite-output interventions; changes in rankings and actions did not imply consistent closed-loop gains.
- Undergraduate Thesis: Personalized Pricing of Customized Car Features** SCNU / Aberdeen
 Advisor: Dr. Binod Bhattacharai; thesis grade: 92/100 Jan. 2026 – Apr. 2026
- Formulated multi-round pricing under hidden buyer preferences as a finite-horizon POMDP, separating observable profiles and negotiation history from latent valuation and bargaining traits.
 - Designed a heterogeneous consumer simulator with dynamic willingness to pay and a frozen bank of 50,000 personas across fixed training, validation, and test splits.
 - Built a persistent negotiation environment; implemented heuristic, PPO, and DreamerV3 seller agents and evaluated them on 5,000 matched test episodes.

- Found that deal rate and profit diverged in this benchmark: the concession heuristic led in average profit, while DreamerV3+CLIP had the highest deal rate.
- Ablated CLIP-based observations and reward shaping, and evaluated a limited inference-time reranking extension.

Draft-and-Target Sampling for Video Generation Policy

SCNU

Second Author, manuscript under review

Apr. 2025 – Sep. 2025

- Proposed **Draft-and-Target Sampling (DTS)**, a training-free acceleration framework for diffusion-based video generation that uses dual denoising trajectories within a single model.
- Introduced **Token Chunking** and **Progressive Acceptance** to reduce redundant resampling while preserving output fidelity.
- Achieved up to **2.1× speed-up** on *Meta-World*, *iThor*, *Libero* with minimal accuracy drop, and surpassed baselines in accuracy on *iThor*.
- Studied the method in embodied-style video-policy settings, emphasizing inference efficiency for downstream decision-making rather than only generic video generation quality.
- **Personal Contributions:** Implemented and refined the core pipeline, ran ablations and benchmarks on *Meta-World* and *iThor*, organized results, created task visualizations, and contributed to manuscript writing and revision.

Test Time Training for 4D Medical Image Interpolation

SCNU

Co-first Author, paper published in the Proceedings of IJCNN 2025 (Oral)

Sep. 2024 – Feb. 2025

- Proposed a **Test Time Training (TTT)** framework for 4D medical image interpolation to handle distribution shift without requiring target-domain labels.
- Designed a shared-encoder adaptation mechanism using self-supervised tasks, including *rotation prediction* and *3D-MAE reconstruction*, during inference.
- Achieved state-of-the-art results on **Cardiac MRI** and **4D-Lung CT (33.7dB and 34.0dB PSNR)** with improved robustness under distribution shift.
- Framed the method as a reusable adaptation template for downstream medical vision tasks such as segmentation and registration under shifting test distributions.
- **Personal Contributions:** Implemented the core 3D-MAE module, ran experiments and metric evaluation, created visualizations, improved code clarity, and contributed to manuscript writing and revision.

Honors & Awards

SCNU Outstanding Undergraduate Thesis (Top ~3%)	Jun 2026
Selected among the top undergraduate theses in the graduating cohort	
SCNU Comprehensive Scholarship (3rd Prize)	Apr 2026
Merit-based award recognizing strong academic record and overall performance	
SCNU Outstanding Student Scholarship	Mar 2025
Awarded to top students with excellent academic performance (Top 8% in cohort)	
SCNU-UoA Joint Institute Scholarship (3rd Prize)	Oct 2024
Granted for outstanding GPA performance, ranked 5th out of 86 students	

Selected Course Project

Self-Supervised Image Classification: SimCLR vs. MAE	2025
<i>Aberdeen Institute, SCNU</i>	
Compared SimCLR and MAE with ViT-Base encoders on ImageNette using the same data split; evaluated downstream accuracy and GPU memory in a single-seed study.	

Skills & Qualifications

- **Programming & ML:** Python, PyTorch, Hugging Face, Transformers, diffusion models, reinforcement learning, scikit-learn.
- **Research Tools:** Git, LaTeX, Jupyter, NumPy, pandas, Matplotlib, Seaborn, Zotero.
- **Tests & Languages:** IELTS Academic 7.5 (L 8.5, R 8.5, W 6.5, S 7.0); GRE 319 (Q 170, V 149, AW 3.5); Mandarin and Cantonese (Native), English (Fluent).